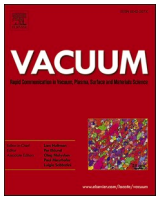
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Suppression of defect formation in atomic-layer deposited Al_2O_3 thin films by addition of AlF_3 cycles

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ABSTRACT

Aluminum oxyfluoride (AlO_xF_y) nanolaminate thin films with various compositions were grown by alternating atomic layer deposition (ALD) cycles of aluminum oxide (Al_2O_3) and aluminum fluoride (AlF_3) at different ratios. The addition of the AlF_3 cycles significantly reduced the defects in the ALD-grown Al_2O_3 films. The defects inside the AlO_xF_y nanolaminate films were analyzed by exposing the defects inside the films by plasma etching after depositing them on a silicon (Si) substrate so that the etching gases penetrating through these defects etch the Si substrate and leave the etching traces. It was confirmed that the defect density inside the AlO_xF_y films was significantly reduced when the ALD cycle ratio of Al_2O_3 and AlF_3 was optimal. These results were experimentally verified by observing the changes in the characteristics of indium-gallium-zinc-oxide (IGZO) thin-film transistors (TFTs) when AlO_xF_y films were covered over the TFTs and hydrogen gas was allowed to permeate the films for a certain period of time. When the ALD cycle ratio of Al_2O_3 and AlF_3 was 5:1, hydrogen penetration was the lowest and the change in characteristics of the IGZO TFTs was also the smallest.

1. Introduction

The term atomic layer deposition (ALD) has been used around the year 2000 and is now actively applied in many semiconductor manufacturing processes [1]. This ALD is a deposition method that builds up single-atomic layers from the substrate surface through self-limited chemical reactions of precursor materials. Due to this principle, ALD has the advantage of being able to deposit thin films with more uniform thickness and better step coverage than any other existing deposition methods. These advantages make ALD essential for applications where film uniformity or step coverage is very important, such as dielectric layers for transistors or diffusion barriers. ALD can be applied to deposit various thin films, but among them, aluminum oxide (Al_2O_3) thin films have been most actively studied due to their chemical and thermal stability, excellent dielectric properties, and good adhesion to various materials.

Although thin films deposited by ALD are often described as defect-free or pinhole-free, but strictly speaking, intrinsic defects or pinholes may inevitably exist depending on the substrate conditions. The mechanism of such intrinsic defects generated in the Al_2O_3 ALD process using trimethyl aluminum (TMA) and water as precursors has been reported in detail in the literature [2]. The fundamental cause of defects may be the imperfect chemical bonding between the functional groups on the substrate and the precursors, and the imperfect bonding is also related to the molecular structure of the precursors. Various types of defects, such as pinholes, point defects, or line defects can occur not only due to the molecular structures of the precursors but also due to the randomness of the surface reactions. The defect density in Al_2O_3 thin films grown by ALD has been typically measured by electroplating or etching in the liquid phase. When copper is electroplated on very thin Al_2O_3 films deposited on metal substrates by ALD, copper grows through the defects in the Al_2O_3 thin films, and the defect density of the Al_2O_3 surface can be measured by examining the traces of the plated copper on the surface [2–8]. Alternatively, the defect density can be determined by etching the metal beneath the Al_2O_3 layer with a liquid etchant that penetrates the defects within the Al_2O_3 layer [8,9]. Another unique method to measure the defect density in Al_2O_3 ALD layers is to supply fluorescent marker molecules through the defects and selectively bind them to the substrate, so that defects with a diameter as small as 200 nm can be identified by their fluorescence [10]. All of the above-mentioned previous studies were performed by electroplating, chemical etching, or fluorescent tracer solutions in the liquid phase to penetrate the defects in the Al_2O_3

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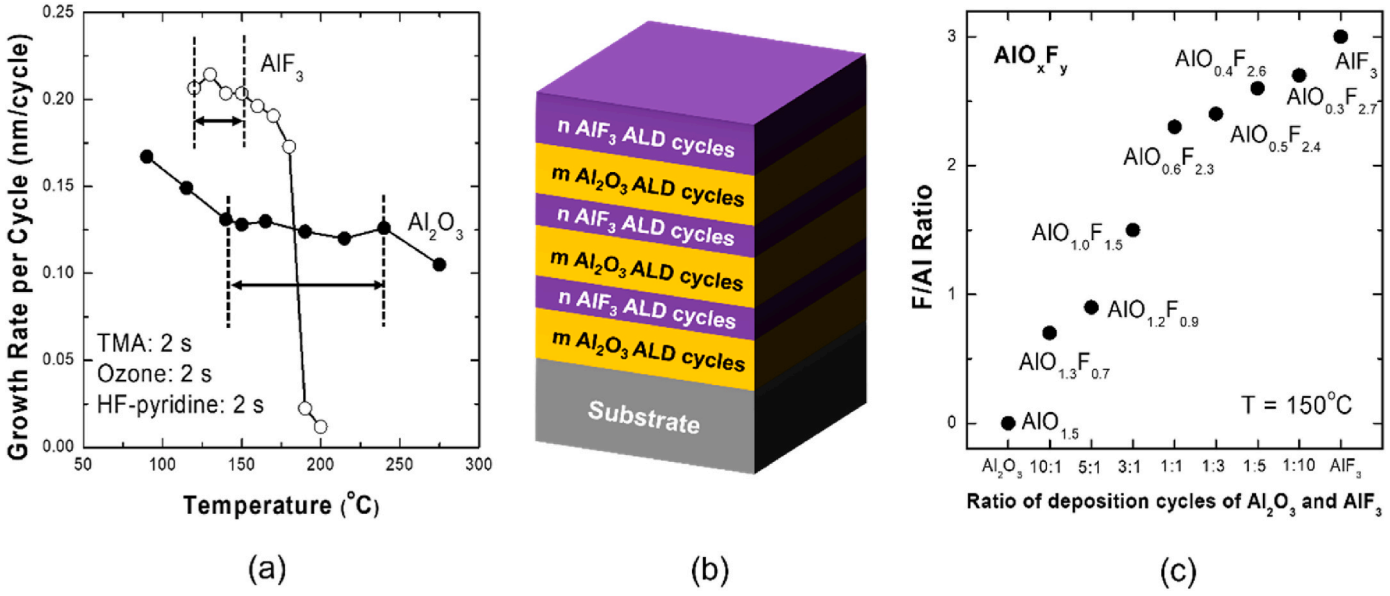


Fig. 1. (a) Temperature windows for ALD of Al_2O_3 and AlF_3 thin films, (b) structure of AlO_xF_y nanolaminate films, and (c) compositions of AlO_xF_y nanolaminates according to the ratio of Al_2O_3 and AlF_3 ALD cycle numbers.

thin films, which limited the size of the defects that can be identified and the types of substrates that can be used for the experiments. To overcome these shortcomings, a novel method was developed by exposing a plastic substrate on which an Al_2O_3 ALD layer was deposited to oxygen plasma, so that gaseous oxygen radicals that penetrated the defects in the Al_2O_3 layer would etch the underlying plastic substrate, thereby measuring the defect density [11]. This study made a great advance by using molecules in the gas phase to measure the defect density in Al_2O_3 thin films and simulating the distribution using a probabilistic cluster model. However, the identifiable defects in Al_2O_3 thin films are limited to pinholes connecting from the surface to the plastic substrate, and it is still difficult to identify isolated defects inside the thin film. In this study, plasma etching was performed using an etching gas capable of etching Al_2O_3 thin films at a very low rate, which enabled the identification of not only pinholes but also isolated defects inside the thin film.

Meanwhile, it has been reported that a nanolaminate structure formed by alternately depositing metal oxide films such as Al_2O_3 grown by ALD and other types of metal oxide films can conceal defects in the Al_2O_3 film and reduce gas diffusion [12–15]. This can be interpreted as the fact that the structural complexity increases because different metal precursors have different bonding numbers with oxygen atoms, which lowers the probability of pinholes or defects occurring. Many types of nanolaminates grown by adding SiO_2, ZnO, TiO_2, ZrO_2, Ta_2O_5, La_2O_3, Y_2O_3, and HfO_2 to Al_2O_3 by ALD have been reported [12]. In this study, unlike previous studies, we investigated whether the formation of defects in Al_2O_3 films could be suppressed by using a nanolaminate structure in which oxygen is partially replaced by fluorine instead of changing the metal precursor. Aluminum fluoride (AlF_3) thin films can be grown by ALD and have been studied to improve the poor corrosion resistance of Al_2O_3 in a fluoride-based plasma environment [16,17]. Aluminum oxyfluoride (AlO_xF_y) thin films formed by nanolamination of Al_2O_3 and AlF_3 can be applied not only as a passivation film to improve plasma stability but also as an optical coating utilizing the low refractive index of AlF_3 [18]. To verify whether adding AlF_3 cycles to the layer-by-layer deposition of Al_2O_3 ALD sequence has a defect suppression effect in Al_2O_3 films, AlO_xF_y ALD thin films were deposited on the surface of silicon (Si) wafers and plasma etched. The plasma etching gas penetrates through the pinholes in the AlO_xF_y layer to etch the Si substrate. During this process, if the plasma gas etches the AlO_xF_y layer at a very low rate, isolated defects existing inside the layer can be gradually exposed to the surface. In this case, the Si substrate can be etched even through the isolated defects exposed to the surface, so it is possible to confirm the isolated defects inside the AlO_xF_y layer. In this experiment, we confirmed that adding AlF_3 cycles can significantly reduce the defect density in the Al_2O_3 thin film grown by ALD. To further verify this reduction in the defect density in the film, the hydrogen barrier properties of the AlO_xF_y film were evaluated. Thin film transistors (TFTs) using oxide semiconductors such as indium-gallium-zinc-oxide (IGZO) are greatly affected by hydrogen penetration on the transistor characteristics. Therefore, when the AlO_xF_y thin film was deposited on top of an IGZO TFT and then exposed to hydrogen plasma, the characteristics of the TFT change more severely as the defect density of the AlO_xF_y thin film increases. These two experiments showed consistent results, concluding that the addition of AlF_3 cycles can suppress defect formation in Al_2O_3 ALD films.

2. Experimental

Trimethyl aluminum (TMA), ozone, and HF-pyridine were used as precursors for ALD of Al_2O_3 and AlF_3 films. The minimum injection time for saturation of the precursors was determined by measuring the deposition rate per cycle (GPC) while increasing the injection time of the precursors. The injection time was determined to be 2 s for all three precursors, and the purge was performed using Argon gas for 20 s until the pressure inside the chamber became 0.01 torr. In order to determine the ALD window of the Al_2O_3 and AlF_3 films, GPC was measured while changing the deposition temperature from 80 to 280 °C. The measured GPCs were 1.25 Å/cycle and 1.8 Å/cycle, respectively. In order to form the AlO_xF_y nanolaminate film by alternately growing the two films at the same temperature, the deposition temperature was determined to be 150 °C, which overlaps the ALD windows of the two films. The composition of AlO_xF_y nanolaminate films was varied by controlling the cycle ratio of Al_2O_3 and AlF_3. The nanolaminate composition was measured with X-ray photoelectron spectroscopy (XPS, K-alpha plus, Thermo Fisher Scientific). The X-ray source was an Al anode providing 1486.6 eV photons. The energy analyzer was calibrated using reference Cu, Ag, and Au samples. The samples were cleaned by etching with Ar for 5 s.

To identify defects inside the grown AlO_xF_y ALD thin films, the thin films were grown on the surface of Si wafers to a thickness of 50 nm and then plasma etched for 20 min. The plasma etching was performed at plasma powers of 100 W and 150 W using a mixture of 30 sccm of CF_4

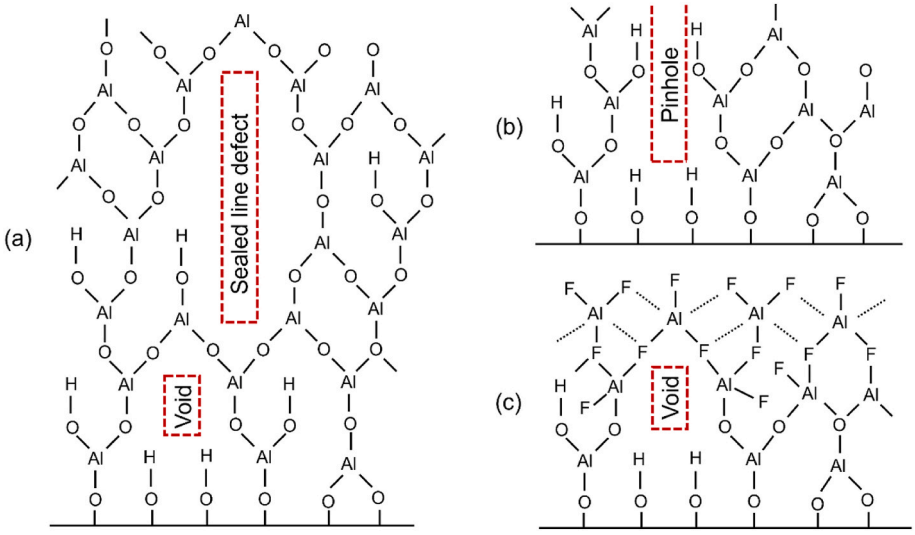


Fig. 2. Creation of (a) point defects (voids), sealed line defects, and (b) pinhole defects in Al_2O_3 ALD growth, (c) suppression of defect creation through introduction of AlF_3 cycles in Al_2O_3 ALD growth.

and 10 sccm of oxygen in a reactive ion etching chamber. During the plasma etching process, the pressure inside the chamber was maintained at 0.19 torr. The etching gases penetrated into the Si substrate along the defects exposed as the thin film surface was etched, leaving etching traces on the surface of the Si substrate. These etching traces were observed using a scanning electron microscope (SEM) to identify defects inside the AlO_xF_y ALD thin films.

To fabricate IGZO TFTs, an IGZO thin film was magnetron sputtered on top of a p-type Si wafer on which a 150 nm thick thermal oxide film was deposited. The thickness of the IGZO thin film was 20 nm, and it was annealed at 300 °C for 1 h after deposition. Finally, aluminum source and drain electrodes were thermally evaporated to fabricate IGZO TFTs. On top of the fabricated IGZO TFT, an AlO_xF_y nanolaminate film was deposited with a thickness of 50 nm by ALD. In order to confirm the permeation of hydrogen through this nanolaminate film, the temperature of the TFT device was maintained at 80 °C, and it was exposed to hydrogen plasma generated with a power of 30 W for a certain period of time. The TFT devices were exposed to hydrogen plasma at 120 s intervals for up to 600 s. The transfer curves of the IGZO TFTs exposed to hydrogen plasma were measured at a drain voltage of 1 V and compared to confirm the effect of the permeated hydrogen.

3. Results and discussion

3.1. Atomic layer deposition of Al_2O_3, AlF_3 and the nanolaminate thin films

HF-pyridine [19–21] anhydrous HF [22], or SF_6 [23,24] can be used as the F source for ALD of AlF_3 thin films. In this study, HF-pyridine, which is safer than anhydrous HF and does not require plasma utilization, was used. Al_2O_3 and AlF_3 thin films were grown in the same ALD apparatus using TMA, ozone, and HF-pyridine precursors. Ozone is known to leave carbon-related impurities in Al_2O_3 ALD films and cause hysteresis in TFTs [25]; however, in this study, ozone was used as an oxidizing agent to minimize the effect of hydrogen that may remain in the film when H_2O is used as an oxidizing agent. XPS analysis results of Al_2O_3 thin films deposited using ozone as an oxidizing agent showed that there were almost no carbon impurities (Fig. S1). The ALD window, which is the temperature range over which the GPC is constant, was 130–240 °C and 120–150 °C for Al_2O_3 and AlF_3 thin films, respectively, as shown in Fig. 1a. In the case of ALD of AlF_3, GPC appears to gradually decrease from above 130 °C, but the decrease is minimal up to 150 °C, so the ALD temperature window was set from 120 to 150 °C. The GPC of the Al_2O_3 and AlF_3 thin films were 1.25 Å/cycle and 1.8 Å/cycle, respectively. Unlike Al_2O_3, the GPC of AlF_3 began to decrease rapidly when the deposition temperature was above 150 °C, and decreased to almost 0 at about 200 °C. This is because AlF_3 produces various volatile Al products through ligand-exchange reaction with $Al(CH_3)_3$. These reactions occur at room temperature [20], but they become active around 150 °C, and the GPC of AlF_3 begins to decrease rapidly [21]. In order to deposit AlO_xF_y nanolaminate thin films with Al_2O_3 and AlF_3 ALD layers under the same growth conditions, the substrate temperature was set to 150 °C, which overlaps the ALD windows of the two films. The component ratios of Al, O, and F in the nanolaminate films, as shown in Fig. 1b, were controlled by varying the ratio of the Al_2O_3 and AlF_3 ALD cycle numbers. AlO_xF_y nanolaminate thin films were grown by varying the ratio of the number of ALD cycles of Al_2O_3 and AlF_3, i.e., m:n, to 10:1, 5:1, 3:1, 1:1, 1:3, 1:5, and 1:10, and the compositions of the thin films were analyzed by XPS. The XPS depth-profile analysis results and F/Al ratios of each nanolaminate thin film are shown in Fig. S2 and Fig. 1c, respectively. The compositions of AlO_xF_y nanolaminates obtained in this study were almost similar to those reported in the literature [16].

3.2. Defect density of ALD-grown AlO_xF_y thin films

Al_2O_3 thin films grown by ALD are known to have fewer defects than films grown by other deposition methods, but in reality, various types of defects can be generated inside the films due to incomplete chemical reactions during the ALD process. As shown in Fig. 2a and b, various defects such as voids, sealed line defects, and pinholes can exist in the Al_2O_3 thin film. It is known that the residual OH ligands play an important role in the generation of these defects, and the generation mechanism of defects has been reported in the literature [2]. In the ideal case of Al_2O_3 ALD, TMA and OH react completely, so that no OH ligands remain. However, if the OH ligands remain in the film, they act as defect centers. If these defects are covered during the next few ALD cycles, voids are formed, and if they are not covered even after many subsequent ALD cycles, line defects can be formed. If the line defects are eventually sealed after many subsequent ALD cycles, sealed line defects occurs, and if they are not sealed until the ALD cycles are completed, they become pinhole defects. The growth of line defects can be stopped by reducing the thickness of the Al_2O_3 ALD film and forming nanolaminates with metal oxide films having a different bonding number with oxygen. It has been reported that the nanolaminate structures of Al_2O_3 and other metal oxides can suppress the occurrence of defects in

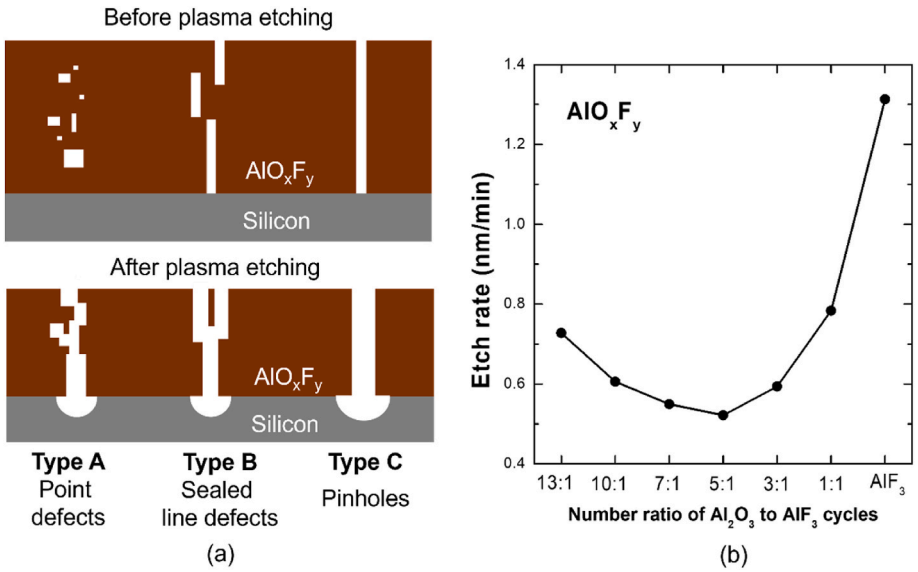


Fig. 3. (a) Schematic diagram showing point defects, sealed line defects, and pinhole defects in AlO_xF_y thin films before and after plasma etching, (b) etching rate of AlO_xF_y thin films according to the ratio of ALD cycle numbers of Al_2O_3 and AlF_3.

the Al_2O_3 film and prevent the penetration of gases such as moisture vapor [12–15]. In this study, AlF_3 thin films, which are oxygen-free, were introduced instead of metal oxides to suppress the growth of defects in Al_2O_3 thin films. Unlike metal oxide nanolaminates in which metal precursors are bonded with OH ligands, in AlO_xF_y nanolaminates, F is directly bonded to Al as shown in Fig. 2c. AlF_3 is represented as shown in the figure because F ions form multi-fold coordination with Al centers, such as five-fold or six-fold. Although the figure may not be accurate because it is difficult to clearly represent the bridging of Al and F at the Al_2O_3 and AlF_3 interfaces, it is expected that the bonds of Al and F can cover the defects in Al_2O_3 thin films caused by the residual OH ligands or suppress their continuous growth into line defects.

From the trend of the F/Al ratio of AlO_xF_y according to the cycle number ratio of Al_2O_3 and AlF_3 shown in Fig. 1c, information about the Al-F bonding state at the interface can be inferred. The figure shows that even when the cycle ratio of Al_2O_3 and AlF_3 is 10:1, the F/Al ratio is very high at 0.7. There can be two possibilities for this high composition ratio of F in AlO_xF_y. One is the possibility of an increase in F incorporation due to fluorination of Al_2O_3 during the HF exposure step of the AlF_3 deposition cycle, and the other is the possibility of an increase in the number of bonds of F with Al at the Al_2O_3 and AlF_3 interface. When AlO_xF_y nanolaminates are deposited by ideal ALD, the F/Al ratio is as shown in

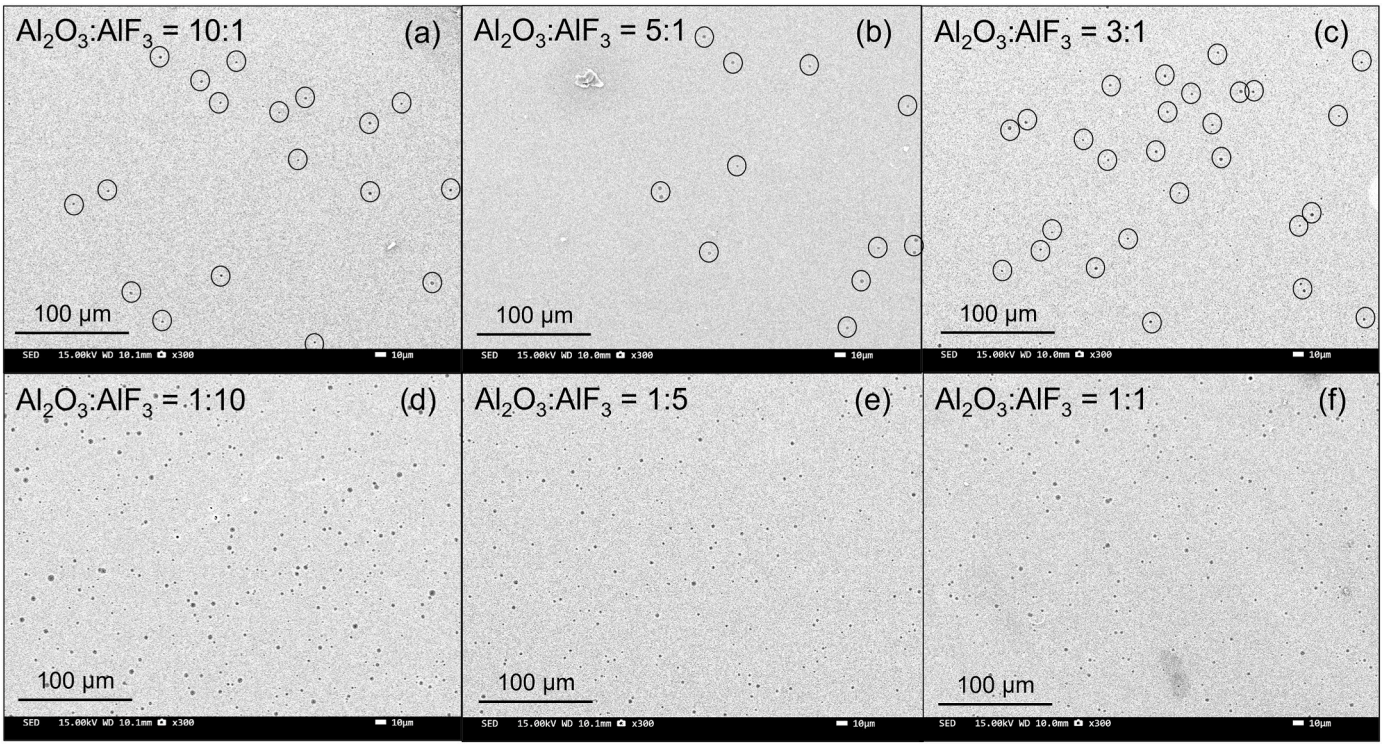


Fig. 4. SEM images of etching traces on the Si substrate surface after plasma etching of AlO_xF_y nanolaminate thin films grown at different ratio of Al_2O_3:AlF_3 cycles. Plasma etching was performed at 100 W for 20 min using CF_4 and O_2 gases.

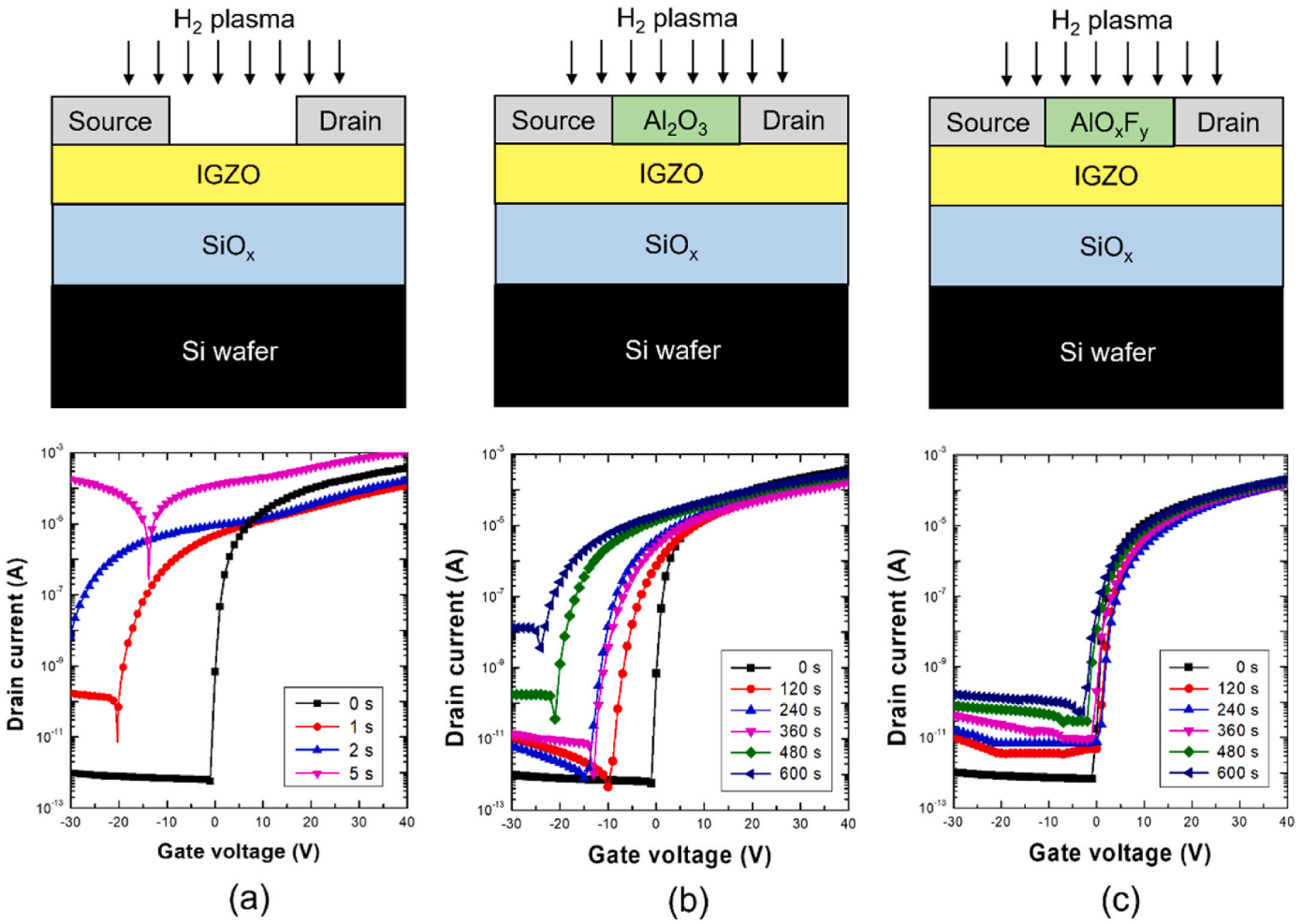


Fig. 5. Changes in electrical characteristics, i.e., transfer curves, of IGZO TFTs over time when exposed to hydrogen plasma in case that (a) the IGZO layer is directly exposed and (b) an Al_2O_3 thin film is covered on top of the IGZO layer. The transfer curves of IGZO TFTs were measured at a drain voltage of 1 V.

Fig. S3. As mentioned, when the number of AlF_3 cycles is smaller than that of the Al_2O_3 cycles, the experimentally measured F/Al ratio is much higher, but in the opposite case, the ratio is almost similar to the ideal case. If Al_2O_3 is fluorinated by HF, the O/Al ratio is expected to decrease compared to the ideal ALD case as O is replaced by F, but the actual experimentally measured O/Al ratio is almost similar to that of the ideal case over the entire region as shown in Fig. S4. This fact suggests that the high F/Al ratio in the AlO_xF_y nanolaminates is due to the increase in the number of bonds of F to Al at the interface of Al_2O_3 and AlF_3 rather than the increase in F incorporation due to fluorination of Al_2O_3. AlF_3 is an ionic solid and can exist in the form of an ionic lattice like AlF_{3+n}^{-n}. To confirm this, when one AlF_3 deposition cycle is inserted between Al_2O_3 layers and AlF_6^{-3} or AlF_4^{-1} ions are formed at the interface, the expected F/Al ratio is as shown in Fig. S5.

Since it is difficult to directly confirm the defects in Al_2O_3 thin films grown by ALD, the defects have been mainly identified by measuring the density of metal traces that appear on the Al_2O_3 surface after electroplating a metal such as copper through pinholes in the Al_2O_3 thin film from a metal substrate [2–8]. However, this method of identifying defects by traces of metal grown along pinholes to the thin film surface does not allow for the detection of point defects or sealed line defects that may exist within the thin film. As shown in Fig. 3a, when plasma etching is performed from the surface of the thin film, defects inside the thin film are gradually exposed, and the exposed defects gradually become thicker and connected by the etching. The etching gases ultimately reach the Si substrate along the connected defects and isotropically etch the substrate surface. Analyzing the etched traces on the Si substrate surface using SEM allows us to identify all defects inside the film, including internal defects, although it is not possible to distinguish all point defects, sealing line defects, and pinhole defects inside the ALD-grown film. The differences between this study and previous studies in analyzing defects in Al_2O_3 thin films grown by ALD are summarized and shown in Table S1. A mixture of CF_4 and O_2 was used as the etching gas so that both Al_2O_3, AlF_3, and Si could be plasma-etched. In order to determine the plasma etching duration, the etching rate according to the AlO_xF_y composition was measured and shown in Fig. 3b. The etching rate varies from a low of 0.5 nm/min to a high of 1.3 nm/min depending on the composition of AlO_xF_y. The etching rate was lowest when the ratio of the number of Al_2O_3 and AlF_3 cycles was 5:1. Considering that the thickness of the AlO_xF_y nanolaminate thin film is 50 nm, the plasma etching duration was set to 20 min in order to etch to a depth of approximately 25 nm from the surface.

The defects that appeared on the surface of the Si substrate after plasma etching of AlO_xF_y nanolaminate thin flms grown by changing the ratio of Al_2O_3 and AlF_3 cycles from 10:1 to 1:10 are shown in Fig. 4. The surface density of etching traces on the substrate caused by the gas penetrating through the defects of the AlO_xF_y thin film were found to decrease when the Al_2O_3:AlF_3 cycle ratio was 5:1 compared to 10:1. These etching traces are indicated by circles in Fig. 4a and b for easy identification. As the cycle ratio of Al_2O_3:AlF_3 further increased, the surface density of etching traces tended to gradually increase. When the number of AlF_3 cycles is greater than that of Al_2O_3 cycles, significantly more defects exist in the AlO_xF_y thin films. These experimental results

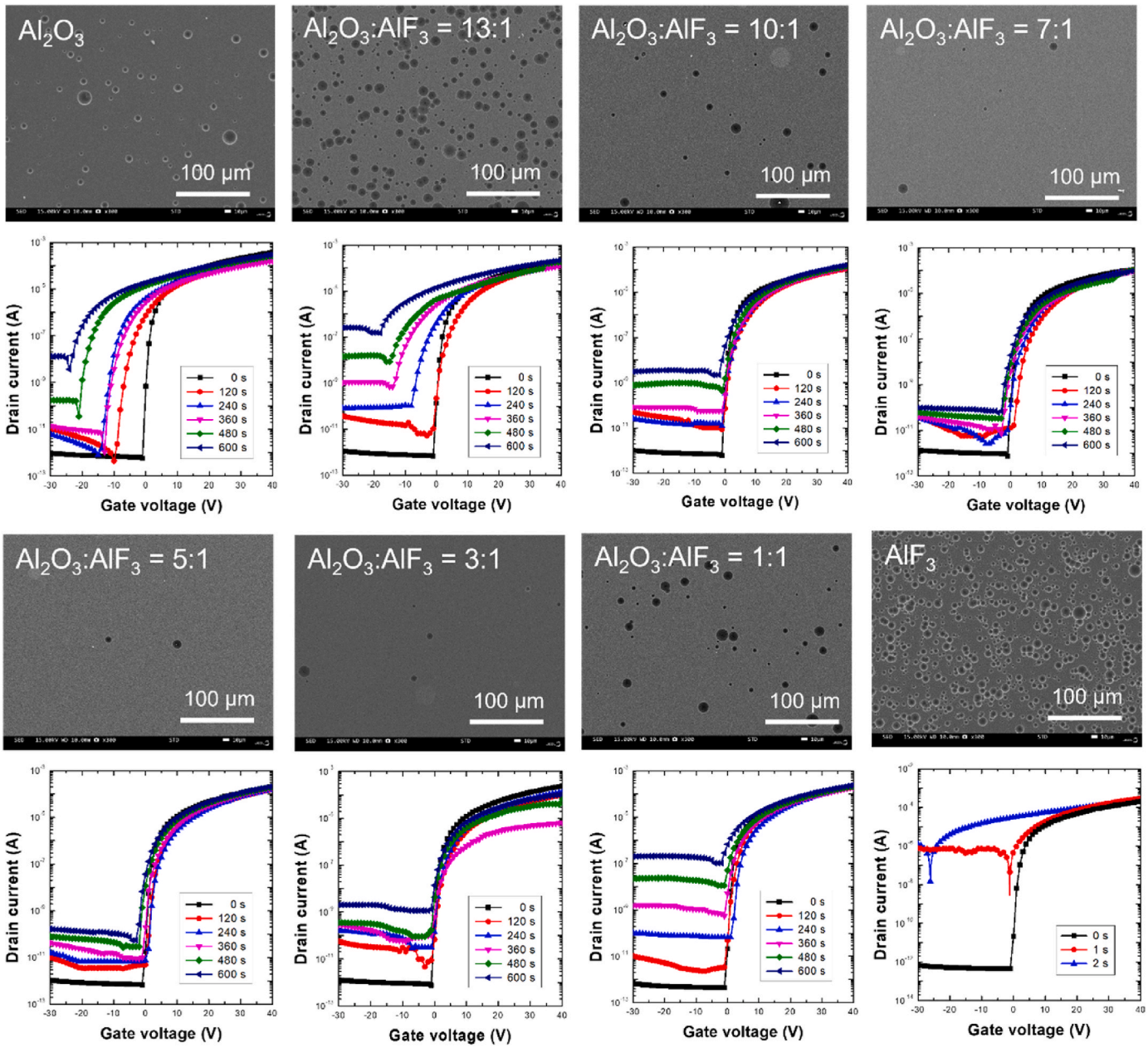


Fig. 6. Defect analysis inside AlO_xF_y nanolaminate films with various Al_2O_3 and AlF_3 cycle ratios and changes in transfer curves of IGZO TFTs with time due to hydrogen penetration through the defects. The transfer curves of IGZO TFTs were measured at a drain voltage of 1 V.

shows that when the ratio of Al_2O_3:AlF_3 cycles is 5:1, the number of defects in the AlO_xF_y thin film is much lower than at other ratios. This shows that, as shown in Fig. 2c, when AlF_3 cycles are introduced at an appropriate ratio during the ALD growth process of an Al_2O_3 thin film, the creation of pinholes or line defects can be suppressed.

3.3. *Verification of AlO_xF_y thin film defects using IGZO TFT characteristic changes due to hydrogen penetration*

Oxide semiconductors such as IGZO, which are mainly used as TFTs for flat panel displays, have their electrical characteristics affected by hydrogen. A small amount of hydrogen improves the electrical characteristics and stability by passivating the oxygen vacancies of the oxide semiconductors, but an excessive amount of hydrogen shifts the threshold voltage (V_{th}) of the TFT in the negative direction and increases the leakage current, ultimately causing the TFT to lose its characteristics [26,27]. In this study, for the purpose of identifying defects within thin films grown by ALD, the thin films were deposited on top of IGZO TFTs and intentionally exposed to hydrogen so that hydrogen that penetrated through the defects in the films induced changes in the electrical characteristics of the IGZO TFTs. Since the electrical characteristics changes become more severe as hydrogen permeates more, the defect density within the thin film can be predicted from this. First, an IGZO TFT as shown in Fig. 5a was fabricated, and then the device was exposed to hydrogen plasma, after which the changes in the TFT characteristics over time were observed. Before exposure to hydrogen plasma, the TFT showed a leakage current less than 1×10^{-12} A at negative gate voltages and a typical switching characteristics in which a drain current flowed at positive gate voltages. However, as expected, after 1 s of exposure to hydrogen plasma, the V_{th} shifted significantly toward the negative voltage, and after 5 s, the switching characteristics were completely lost. This is because the IGZO layer is directly exposed to hydrogen plasma, so a large amount of hydrogen easily penetrates the IGZO surface and rapidly affects the electrical characteristics of the TFT. Even when the

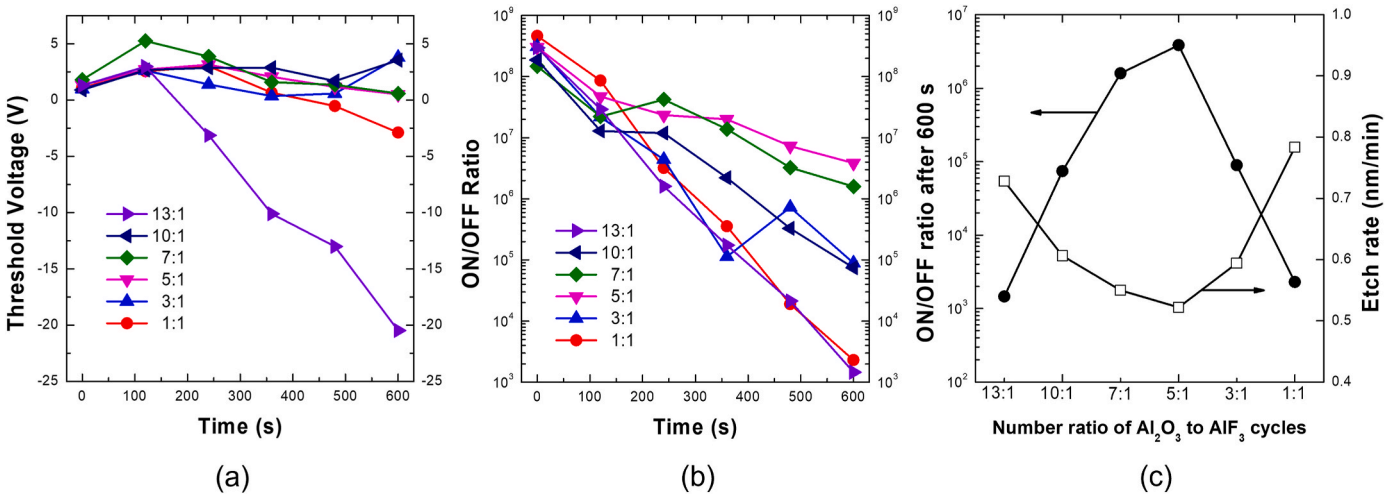


Fig. 7. Changes in (a) V_{th} and (b) on/off ratio of IGZO TFTs covered with AlO_xF_y nanolaminate films according to hydrogen plasma exposure time, (c) etching rate of AlO_xF_y nanolaminate films according to the cycle ratio of Al_2O_3 and AlF_3, and on/off ratio of IGZO TFTs measured after 300 s of hydrogen plasma exposure.

IGZO surface was covered with an Al_2O_3 thin film grown by ALD, as the exposure time to hydrogen plasma increased, hydrogen gradually penetrated, shifting the V_{th} and increasing the leakage current, but the time became considerably longer, as shown in Fig. 5b. This means that there are still many defects in the Al_2O_3 thin film through which hydrogen can penetrate. As shown in Fig. 5c, when AlO_xF_y nanolaminate thin films (the cycle ratio of Al_2O_3 and AlF_3 is 5:1) are deposited on the surface, the change in the transfer curve of the IGZO TFT is significantly reduced even under hydrogen plasma exposure for the same period of time. This result is because the AlO_xF_y nanolaminate thin films prevent hydrogen penetration from the outside, thereby reducing the change in the electrical characteristics of the IGZO TFT.

In the experiment shown in Fig. 4 above, it was shown that the AlO_xF_y nanolaminate films with a greater number of Al_2O_3 cycles than AlF_3 cycles had significantly fewer defects than the opposite case. The results allow us to predict that the defects in the AlO_xF_y films will be the lowest when the ratio of Al_2O_3 and AlF_3 cycles is around 5:1. For verification, the same experiment was performed by further dividing the ratio of Al_2O_3 and AlF_3 cycles. At this time, the power for plasma etching was increased to 150 W to more easily confirm the etching traces on the Si substrate surface. The results are shown in Fig. 6. The AlO_xF_y nanolaminate films were grown by ALD with the cycle ratios of Al_2O_3 and AlF_3 varying from 13:1, 10:1, 7:1, 5:1, 3:1, and 1:1. Fig. 6 shows the etching traces on the Si substrate surface after plasma etching the films for 20 min. In addition, the change in the transfer curve over time when IGZO TFTs with these thin films on the surface were exposed to hydrogen plasma is also shown in the figure. As expected, when the cycle ratio of Al_2O_3 and AlF_3 was 5:1, the lowest defect density was observed on the substrate surface, and the change in the transfer curve according to the time of exposure to hydrogen plasma was also found to be the smallest.

In the case of pure Al_2O_3 thin films and AlO_xF_y nanolaminate films with the cycle ratio of Al_2O_3 and AlF_3 of 13:1, as the hydrogen plasma exposure time increases, the V_{th} shifts toward the negative voltage direction and the off current also increases significantly. These thin films have many defects inside, so they leave a high density of etching traces on the Si surface after plasma etching. However, as the number of AlF_3 cycles increases relatively compared to the number of Al_2O_3 cycles, the defects inside the AlO_xF_y nanolaminate films are significantly reduced, and the lowest internal defect density is observed when the cycle ratio of Al_2O_3 and AlF_3 was 5:1. When the number of AlF_3 cycles increases beyond this ratio, the defect density tends to increase significantly again. When the cycle ratios of Al_2O_3 and AlF_3 are 3:1 and 1:1, the off current increases significantly as the hydrogen plasma exposure time increases. For pure AlF_3 thin films, the defect density within the thin film is so high that the TFT loses its switching characteristics even after only 2 s of hydrogen plasma exposure. The results of identifying defects inside the thin films by examining the etching traces left on the Si surface after plasma etching the AlO_xF_y nanolaminate films were almost similar to the results of hydrogen plasma stability when the same thin films were used as hydrogen barriers for IGZO TFTs. The validity of the defect analysis method inside the thin films performed in this study was verified through the analysis of the electrical characteristics of the IGZO TFT.

The V_{th} and on/off current ratio of IGZO TFTs using AlO_xF_y hydrogen barrier layers with different cycle ratios of Al_2O_3 and AlF_3 are plotted as functions of hydrogen plasma exposure time in Fig. 7. As shown in Fig. 7, the smallest change in V_{th} was observed when the cycle ratio of Al_2O_3 and AlF_3 was 5:1, and the smallest decrease in on/off ratio was also observed in this case. When the ratio of the AlF_3 cycle to the Al_2O_3 cycle was very low, such as 1/13, V_{th} decreased rapidly as the hydrogen plasma exposure time increased, but when the ratio increased, the change in V_{th} was not significant. Unlike V_{th}, the on/off ratio showed a common tendency to decrease as the hydrogen plasma exposure time increased in all cases. It is generally known that V_{th} decreases as hydrogen is incorporated more into the IGZO channel, but in the case where the AlO_xF_y film was in contact with the IGZO channel as in this study, it showed a trend different from the general expectations. This is because, as reported in the literature [28], the fluorine atoms included in the AlO_xF_y film stabilize the V_{th} shift for hydrogen incorporation by fluorination of the IGZO channel. To summarize the above experimental results, when the cycle ratio of Al_2O_3 and AlF_3 was 5:1, the plasma etching rate was the lowest and the defect density inside the AlO_xF_y nanolaminate film was also the lowest. At the same time, the AlO_xF_y nanolaminate film deposited at this cycle ratio most effectively prevented hydrogen penetration, so that the on/off ratio of the IGZO TFT could be maintained at the highest level even after exposure to hydrogen plasma for 600 s. As shown in Fig. 7c, the plasma etching rate of the nanolaminate film itself and the hydrogen barrier effect of the IGZO TFT showed consistent results in that the defect density inside the AlO_xF_y nanolaminate film was the lowest when the cycle ratio of Al_2O_3 and AlF_3 was 5:1.

4. Conclusions

In order to reduce the internal defect density of Al_2O_3 ALD films, AlO_xF_y nanolaminate films were deposited by adding AlF_3 ALD cycles. Considering that the internal defects of Al_2O_3 ALD films are greatly affected by the -OH groups present inside the films, AlF_3 ALD films

without the effect of -OH groups were introduced to form the AlO_xF_y nanolaminate films. In order to analyze the density of defects existing inside the films, the AlO_xF_y films were plasma-etched at a very low speed, and the etching gas penetrating through the exposed defects etched the Si substrate, thereby analyzing the defect density inside the AlO_xF_y films. The experimental results showed that the AlO_xF_y films with a cycle ratio of Al_2O_3 to AlF_3 of 5:1 had the lowest defect density, and this was further verified by observing the change in electrical characteristics due to hydrogen penetration using the AlO_xF_y films as a hydrogen barrier for IGZO TFTs. From these experimental results, it was demonstrated that the addition of AlF_3 ALD cycles could suppress defect formation in Al_2O_3 ALD thin films and that there exists an optimal Al_2O_3 and AlF_3 cycle ratio. The AlO_xF_y nanolaminate thin films with the optimal ratio that can minimize defect density can be utilized as a barrier layer that can prevent the penetration of gases such as moisture or hydrogen, or as a coating film for corrosion prevention.

CRediT authorship contribution statement

Chang-Yun Na: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Changjun No:** Visualization, Methodology, Investigation. **Byungkwon Lim:** Writing – review & editing, Validation, Supervision, Resources, Conceptualization. **Sung Min Cho:** Writing – review & editing, Validation, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.vacuum.2025.114154.

Data availability

Data will be made available on request.

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